

Sensor Signal Analysis for Predicting Semiconductor Manufacturing Defects

Paul J Anderson

Abstract

Semiconductor manufacturing yield depends on early failure detection. This study analyzes the SECOM dataset—1,567 observations from a semiconductor fabrication line, each characterized by 590 anonymized sensor measurements and a binary pass/fail label. The data present three compounding challenges: severe class imbalance (~93% pass), substantial sensor missingness, and a high-dimensional feature space relative to the number of observed failures ($n = 99$). We address two research questions: which sensors are most strongly associated with product failure, and can failure be predicted reliably from sensor readings alone? Following a principled preprocessing pipeline and chronological holdout evaluation, we fit an elastic net-penalized logistic regression with alpha and lambda selected by cross-validated AUC. A single sensor—Sensor 59—is selected by the final model and independently confirmed by Wilcoxon rank-sum test ($p = 2.30 \times 10^{-8}$), achieving a test AUC of 0.72. The convergent identification of one dominant process signal across two independent methods is the central finding, and suggests Sensor 59 as a priority candidate for process engineer investigation.

Introduction

In semiconductor manufacturing, product yield is a critical performance metric. A single fabrication line produces thousands of units across hundreds of tightly controlled process steps—including photolithography, chemical mechanical planarization (CMP), physical and chemical vapor deposition (PVD/CVD), etch, and metrology—each monitored by an array of sensors. These sensors capture parameters such as electrical characteristics (leakage current, sheet resistance, threshold voltage), physical layer properties (film thickness, planarity, step height), process conditions (chamber pressure, gas flow rates, radio-frequency power), and optical measurements (critical dimension, overlay error). When a unit fails quality inspection, identifying which upstream process conditions contributed to that failure is both technically difficult and economically important—catching failures early reduces waste, scrap costs, and time-to-diagnosis for process engineers.

The SECOM dataset (UCI Machine Learning Repository) provides a concrete case for this problem. Collected from a real fabrication line over eleven months in 2008, it contains 1,567 observations and 590 anonymized sensor measurements per unit, along with a binary pass/fail outcome. Although sensor identities are anonymized, the measurement types likely span electrical test results, deposition uniformity metrics, planarization endpoint signals, and etch rate monitors—making the dataset broadly representative of real fab-line data. Its challenges are equally representative: a failure rate of approximately 7% creates severe class imbalance; substantial missing data across many sensors is consistent with intermittent tool downtime or recipe-specific measurement steps; and the feature space is high-dimensional relative to the number of observed failures, with an events-per-variable ratio well below the recommended minimum of 10.

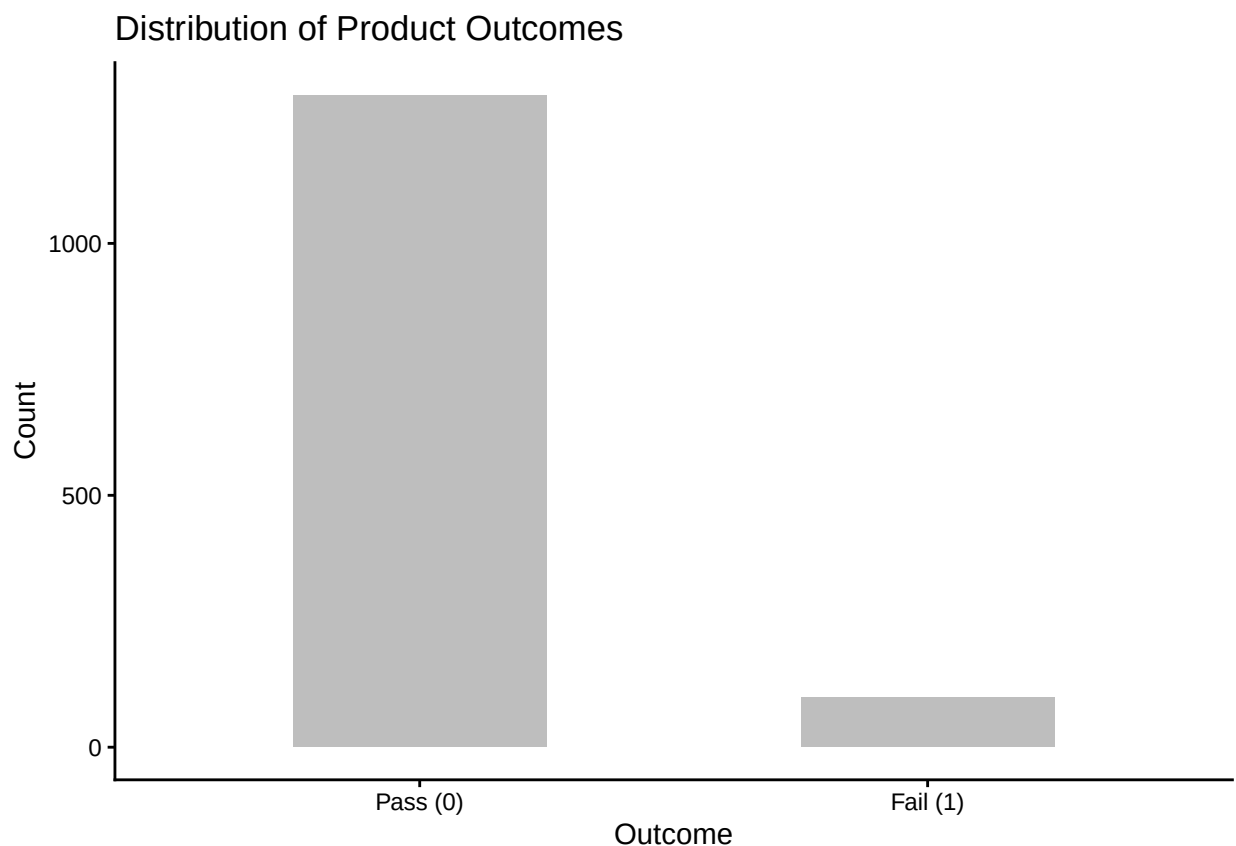
This analysis pursues two goals. The first is explanatory: to identify which sensors show meaningful differences in readings between passed and failed units, providing insight into which process conditions are most associated with failure. The second is predictive: to develop a classification model capable of forecasting product failure from sensor readings, evaluated honestly against the constraints imposed by class imbalance and limited failure events.

Data Processing

We apply a preprocessing pipeline to improve data quality and reduce dimensionality. First, sensors with more than 150 missing values (approximately 10% of observations) are removed, as many measurements are only collected for specific process recipes. Next, rows containing any remaining missing values are dropped to produce a fully observed dataset. The binary pass/fail outcome is then recoded from $\{-1, +1\}$ to $\{0, 1\}$ for compatibility with logistic regression. Sensors with zero variance are removed because they provide no predictive information. Finally, variance-based trimming is applied to address extreme variability across sensors: features outside the 35th–80th percentile of variance are excluded, since very low-variance sensors contribute little information and extremely high variances are likely driven by outliers rather than meaningful process signals. After preprocessing, the final analysis dataset contains 1,393 observations and 190 sensor features, with a failure rate of 7.1% (99 failures and 1,294 passes).

```
## data_clean: 1393 obs, 191 features
```

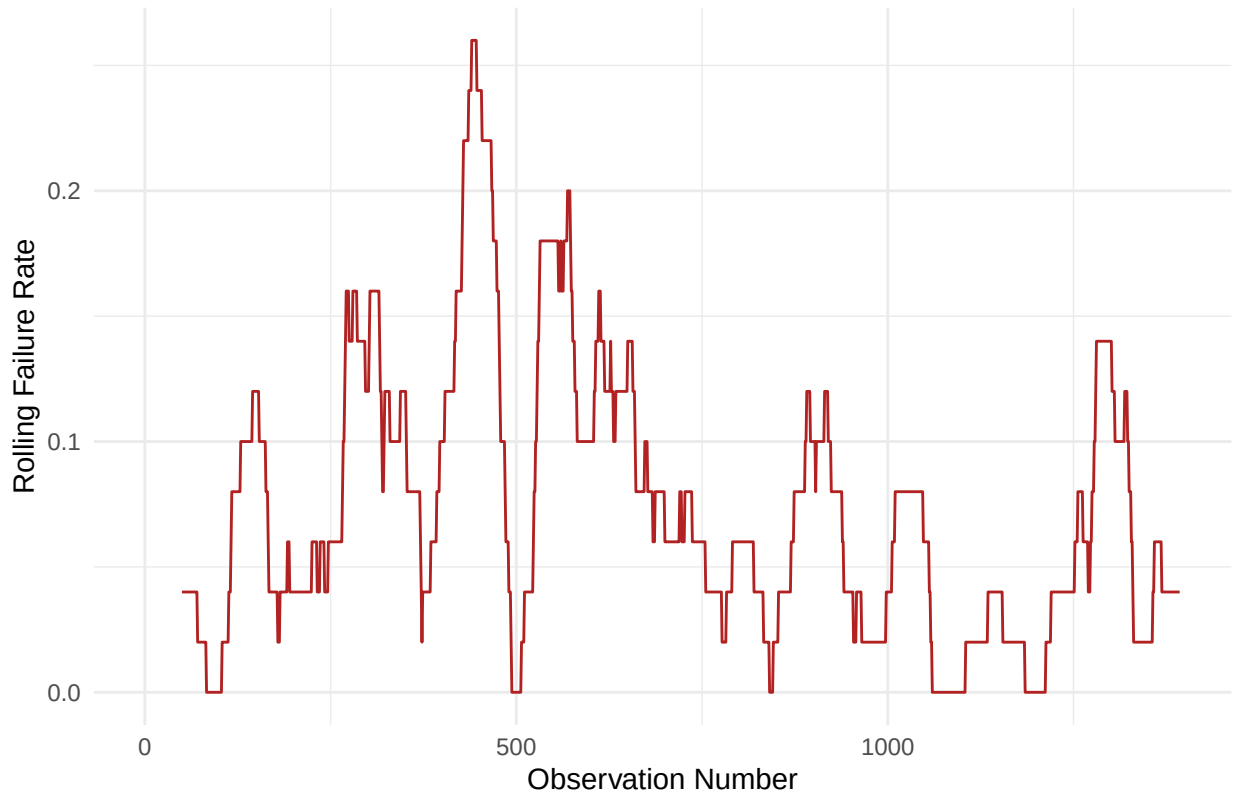
```
## Failures: 99
```



The dataset is severely imbalanced: 1,294 units passed and only 99 failed. This ratio has direct consequences for modeling. A naive classifier that predicts “pass” for every unit achieves 92.9% accuracy while detecting zero failures. Standard model accuracy is therefore uninformative here; we rely instead on AUC, sensitivity, and balanced accuracy instead.

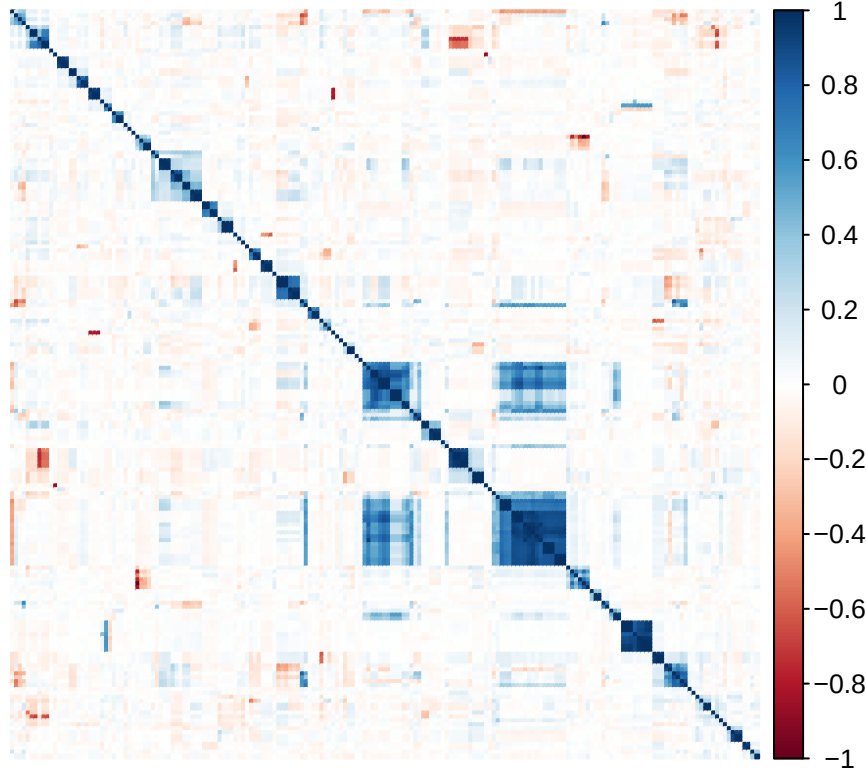
Observations span January 8 to December 10, 2008. Manufacturing processes evolve over time, tool drift, recipe changes, and maintenance events can shift the failure rate non-stationarily. We examine failure rate over time using a rolling 50-observation window.

Rolling Failure Rate Over Observations ($k = 50$)



The spike near observation 450 (timestamp: 2008-08-09 00:32 UTC) is the most prominent feature of the temporal profile. Its sharpness suggests a discrete process event rather than gradual drift. This non-stationarity means observations are not fully exchangeable over time—a violation of the independence assumption in logistic regression. Standard k -fold cross-validation does not account for this. Chronological holdout validation would be more appropriate and is noted as a priority for future work.

Sensor Correlation--All Variables (clustered)



The correlation matrix reveals pronounced block structure among the among all 190 sensors, with several clusters of highly correlated variables. This is consistent with sensors grouped by process stage — for example, multiple planarity or film thickness measurements taken at the same tool step would be expected to co-vary. For modeling, this correlation structure has an important implication: standard logistic regression becomes unstable when predictors are highly collinear. LASSO penalization addresses this directly by selecting one representative from correlated clusters and shrinking the rest toward zero, which is a key justification for the modeling approach taken in Section 3.

Methods & Models

We use a chronological holdout split rather than random partitioning. Random splits allow future observations to leak into training, which artificially inflates performance estimates when the failure process is non-stationary—as is clearly the case here given the pronounced spike near observation 450 (2008-08-09 00:32 UTC).

We initially explored splitting at the spike, training on the pre-anomaly period ($n=450$, 42 failures) and testing on the post-anomaly period ($n=943$, 57 failures). This framing asks a meaningful question—can pre-anomaly sensor patterns generalize to a post-anomaly process—but yielded insufficient training failures for stable model estimation.

We therefore apply an 80/20 chronological split, assigning the first 1,114 observations to training (86 failures, 7.7%) and the remaining 279 to the test set (13 failures, 4.7%). Temporal order is preserved throughout, preventing future data leakage while retaining enough failures for meaningful model estimation.

Training: 1114 obs, 86 failures

Test: 279 obs, 13 failures

We fit an elastic net-penalized logistic regression using `glmnet`, combining L1 and L2 penalties through a

mixing parameter α :

$$\text{Penalty} = \alpha \|\beta\|_1 + \frac{(1 - \alpha)}{2} \|\beta\|_2^2$$

Pure LASSO ($\alpha = 1$) tends to arbitrarily discard sensors from correlated clusters, losing signal from related process stages. Pure ridge ($\alpha = 0$) retains all features, preventing the sparse sensor identification needed to answer the explanatory question. Elastic net balances both. With 86 training failures across 190 features, EPV is well below 1, making penalization essential to control overfitting.

α is selected via grid search over $\{0.25, 0.50, 0.75, 1.00\}$, with λ tuned by cross-validated AUC for each α value. The optimal threshold is selected using Youden’s J statistic on the training ROC curve and applied to the test set to avoid data leakage.

Feature importance is assessed through elastic net coefficient magnitudes and Wilcoxon rank-sum tests on all 190 sensors, with a Bonferroni correction applied to control family-wise error rate.

Results

Grid search over $\alpha \in \{0.25, 0.50, 0.75, 1.00\}$ selected $\alpha = 0.25$, reflecting a predominantly ridge blend with modest L1 sparsity consistent with the block correlation structure observed among sensors. At `lambda.1se`, the model retains a single sensor, Sensor 59, with a positive coefficient of 0.0037. Elevated Sensor 59 readings are associated with increased failure probability. The intercept of -2.50 reflects the low baseline failure rate in the training period.

Table 1: Elastic net coefficients at `lambda.1se` ($\alpha = 0.25$).

Term	Coefficient
Intercept	-2.4958
Sensor 59	0.0037

The model achieves a test AUC of 0.72 on the chronological holdout set. At the Youden-optimal threshold of 0.077, the model correctly identifies 4 of 13 failures (sensitivity = 0.31) while flagging only 14 of 266 passing units as false alarms (specificity = 0.95). Balanced accuracy of 0.63 and kappa of 0.22 both reflect modest but meaningful discrimination above chance. Notably, the less regularized `lambda.min` model—retaining 68 sensors—achieves a lower test AUC of 0.66, confirming that additional sensors introduce noise rather than signal under these low EPV conditions.

Table 2: Model performance on chronological holdout test set (n=279, 13 failures).

Metric	Value
Test AUC	0.7169
Sensitivity	0.3080
Specificity	0.9470
Balanced Accuracy	0.6280
Precision	0.2220
Kappa	0.2160

Wilcoxon rank-sum tests across all 190 sensors identify six sensors with statistically significant distributional differences between passed and failed units after Bonferroni correction ($\alpha = 0.05/190 \sim 0.00026$): Sensors 59, 33, 129, 28, 477, and 205.

Sensor 59 is the dominant signal by a wide margin—its corrected p-value (2.30×10^{-8}) is three orders of magnitude below the next most significant sensor (Sensor 33, corrected $p = 0.010$). The five remaining sensors are near the correction threshold and should be interpreted cautiously given the limited number of failures. The independent convergence of elastic net and Wilcoxon screening on Sensor 59—through entirely different analytical frameworks—is the central finding of this analysis.

Sensors significant after Bonferroni correction: 6 of 191

Table 3: Top 15 sensors by Wilcoxon rank-sum test (all 190 sensors tested, Bonferroni corrected).

	Sensor	p-value	Bonferroni p	Significant
59	59	1.20e-10	2.30e-08	TRUE
33	33	5.30e-05	1.01e-02	TRUE
129	129	7.21e-05	1.38e-02	TRUE
28	28	1.81e-04	3.46e-02	TRUE
477	477	1.92e-04	3.66e-02	TRUE
205	205	2.39e-04	4.56e-02	TRUE
316	316	6.39e-04	1.22e-01	FALSE
341	341	8.80e-04	1.68e-01	FALSE
290	290	1.42e-03	2.71e-01	FALSE
125	125	1.50e-03	2.86e-01	FALSE
122	122	2.35e-03	4.49e-01	FALSE
180	180	4.11e-03	7.85e-01	FALSE
127	127	4.50e-03	8.59e-01	FALSE
336	336	5.16e-03	9.85e-01	FALSE
452	452	6.22e-03	1.00e+00	FALSE

Discussion

Sensor 59 is the dominant signal associated with product failure, identified by both elastic net selection and Wilcoxon testing (corrected $p = 2.30 \times 10^{-8}$). It is the most significant feature by several orders of magnitude and has a positive coefficient, indicating higher values are linked to increased failure risk. Five additional sensors (33, 129, 28, 477, 205) show weaker signals after correction. Without metadata, the physical meaning of Sensor 59 is unknown, but its consistency across methods makes it the primary candidate for process investigation.

Predictively, a single-sensor elastic net model achieves $AUC = 0.72$ on a chronological test set. A larger 68-sensor model performs worse ($AUC = 0.66$), likely due to low event density and added noise. Sensitivity is 0.31 at the Youden threshold, meaning most failures are missed, which is problematic in a high-cost escape setting. Improving recall would substantially increase false positives given the 4.7% failure rate.

The main limitations of the analysis are driven by data scarcity, temporal instability, and limited interpretability. With only 13 failures in the test set, estimates of sensitivity and AUC are inherently unstable and subject to high variance. In addition, the process appears to exhibit temporal drift, as evidenced by differing failure rates between the training period (7.7%) and the test period (4.7%), and a large spike in failures at the 450 mark, which may effect generalization. The lack of semantic information about the sensors further constrains interpretation, since Sensor 59 cannot be linked to a specific physical process without context. Finally, the use of variance-based filtering may have removed potentially informative sensors, introducing additional uncertainty into feature selection.

Given that a single sensor dominates both the predictive and explanatory results, the most impactful next step is identifying the physical process parameter Sensor 59 measures and investigating whether its elevated readings correspond to a known failure mode — tool drift, recipe deviation, or incoming material variation.

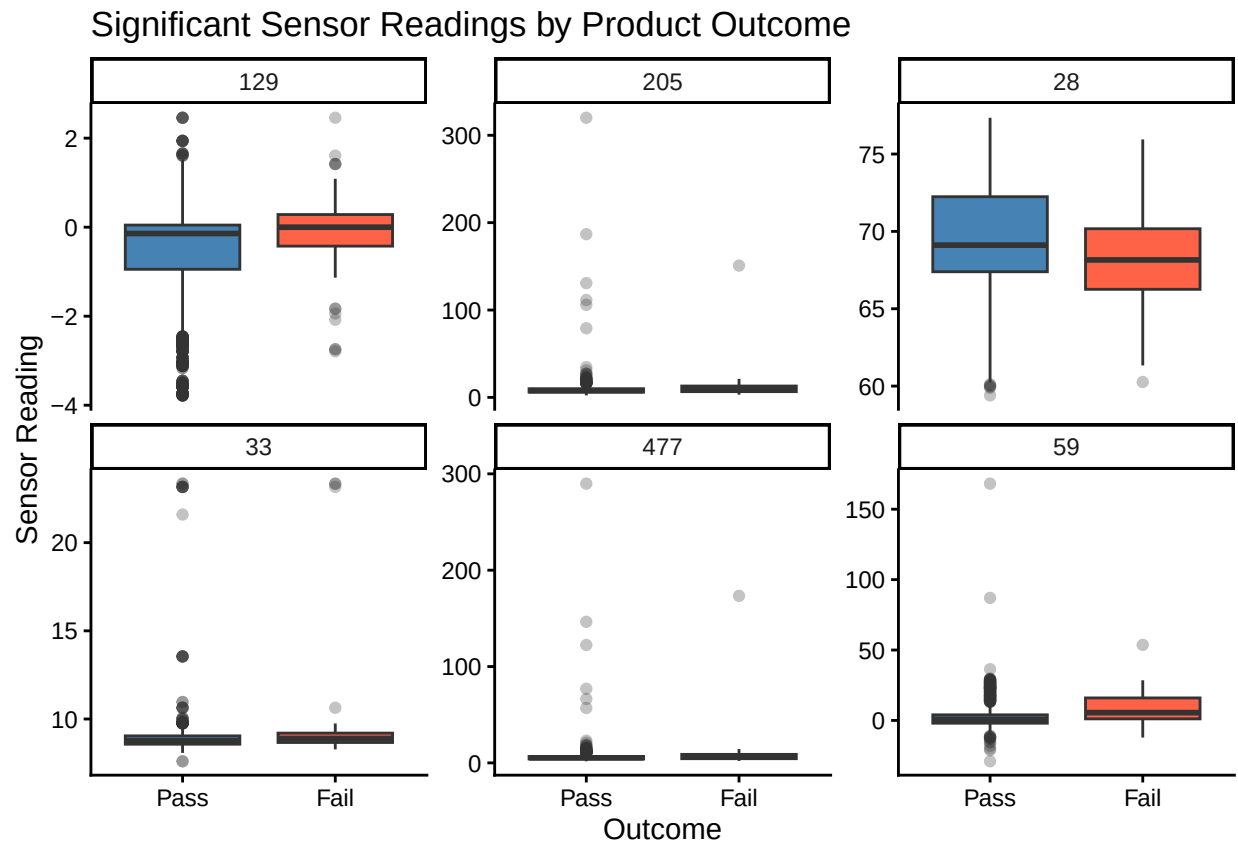


Figure 1: Distributions of the six Bonferroni-significant sensors by product outcome. Sensor 59 shows the most pronounced separation.

Statistically, an anomaly detection approach trained solely on Sensor 59 readings from passing units—such as a control chart or one-class classifier—may be more appropriate and interpretable than a full classification model, and would generalize more naturally to the low-failure-rate manufacturing environment.

Acquiring additional failure data or extending the observation window would improve EPV and enable the spike-based chronological split, which remains a more scientifically motivated evaluation than the 80/20 cut. With sufficient failures, time-series cross-validation may provide more reliable performance estimates than a single holdout period.